Analysis of Experimental and Simulated Test Results for Photonic Integrated Circuits Fabricated Using E-Beam and Photolithography

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***Abstract*—We designed, fabricated and tested two photonic chips-one built using e-beam technology at University of Washington and the second one built at RIT using Photolithography. Both the chips were designed in KLayout using two different Process Design Kits (PDKs). Components from two different libraries-SiEPIC and RIT were used respectively to build the photonic devices on two separate grids. Post fabrication, an automated probe system was employed to test and gather data from photonic devices. The test results obtained using automated test setup is analyzed using MATLAB and Python scripts for further study and research. The compact model for silicon waveguide is developed using Lumerical's Mode Solutions software. Corner Analysis is performed on the waveguide to study the impact of manufacturing variations on the test results. The simulation of the photonic circuits is performed in Interconnect. The experimental data is further compared with simulated results and conclusions are drawn based on our findings.**

***Index Terms*—photonic integrated circuits, silicon photonics, silicon-on-insulator**

I. Introduction

THIS project focuses on analyzing the impact of different arms' length on MZIs performance data. Emphasis is laid on deriving Fast Spectral Range (FSR), group index, and effective index for Mach-Zehnder Interferometers (MZI). Devices using single-bus or double-bus ring resonators are analyzed particularly for their quality factor. Study also focuses on the use of different gaps size for the rings and how it affects the performance of the rings.

II. Waveguide Theory

The silicon waveguide is 220 nm thick and 500 nm wide. These waveguide dimensions are chosen specifically to operate the waveguide in single-mode at 1550 nm wavelength. Unlike an optical fiber for which the critical angle is quite large, the critical angle for silicon waveguide surrounded by glass (SiO_2) is only 24.85°; hence it is possible to incorporate tight bends in the photonic integrated circuits and still experience total internal reflection. The test wavelengths in the laser are set from 1500 nm to 1600 nm.

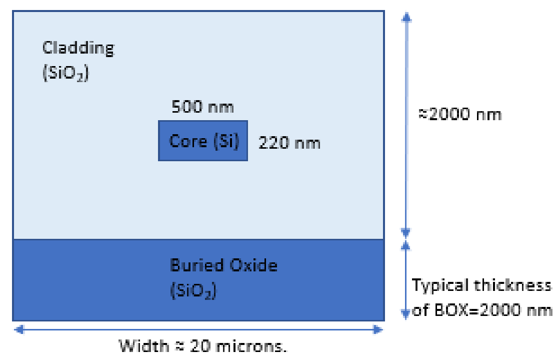


Fig. 1. Typical Silicon Waveguide Structure

It is well-known that silicon by itself does not produce light because of indirect band gap; hence external lasers along with grating couplers are used to guide light through the waveguides. Since silicon waveguides are very small compared to optical fiber; the coupling challenge here is to allow light to travel in and out of the waveguide from/to the optical fiber [1].

The properties that define an optical waveguide mode in a 1D slab waveguide are thickness of the waveguide, material used in making the waveguide, effective refractive index of waveguide and wavelength of light. Fig. 2 and Fig. 3 show simulation results of one-dimensional (1D) mode profile for 220 nm thick slab waveguide at 1550 nm wavelength. The refractive index of glass is chosen to be 1.444 and refractive index of silicon is selected to be 3.47.

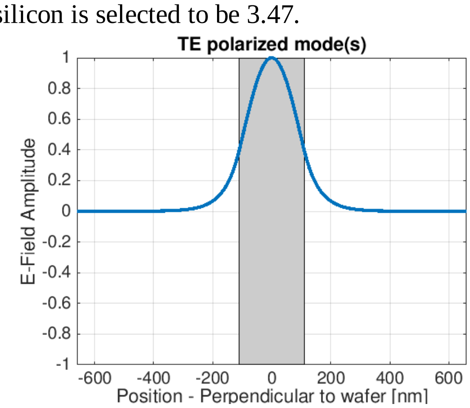


Fig. 2. 1D mode profile of a slab waveguide (TE)

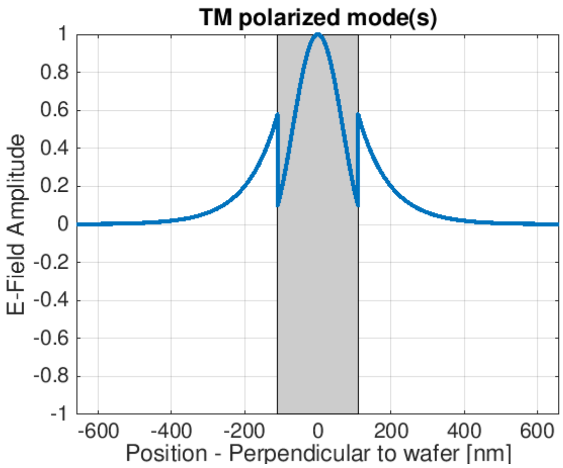


Fig. 3. 1D mode profile of a slab waveguide (TM)

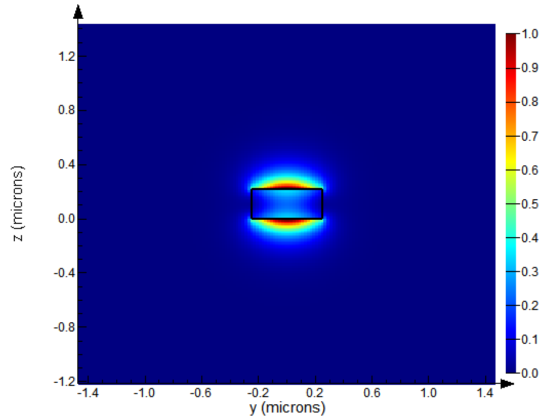


Fig. 5. TM Mode

It can be observed that some discontinuity is observed in the TM mode due to Maxwell's boundary conditions. TM is polarized perpendicular to the silicon-glass interface and the normal component (E_x) of the electric field is discontinuous at the boundary. The grey area indicates the silicon waveguide thickness.

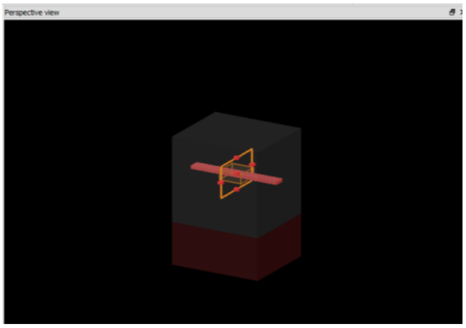


Fig. 6. Perspective View of Simulated Mode Waveguide

TABLE I
EFFECTIVE INDEX VALUE FOR DIFFERENT MODES AT 1.55 μ (1D)

MODE	Effective Index
100% TE mode	2.84185
100% TM mode	2.04889

TABLE II
EFFECTIVE INDEX VALUE FOR DIFFERENT MODES AT 1.55 μ (2D)

MODE	Effective Index
TE mode	2.448277
TM mode	1.774354

III. MODELLING AND SIMULATION

Unlike 1D slab waveguide where mode is entirely TE or TM, mode in 2D slab waveguide simulation is never pure TE or TM.

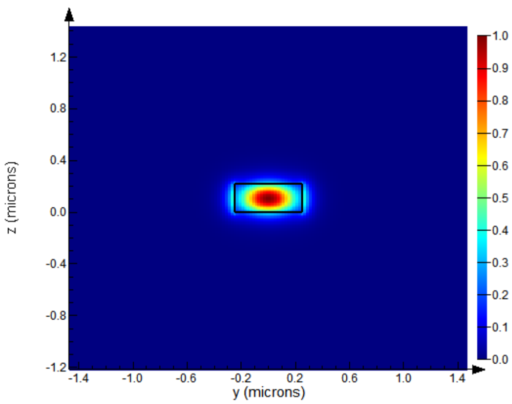


Fig. 4. TE Mode

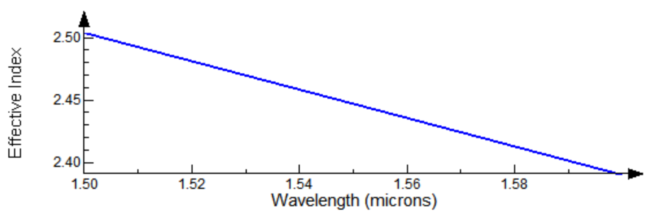


Fig. 7. Effective Index vs. Wavelength (TE)

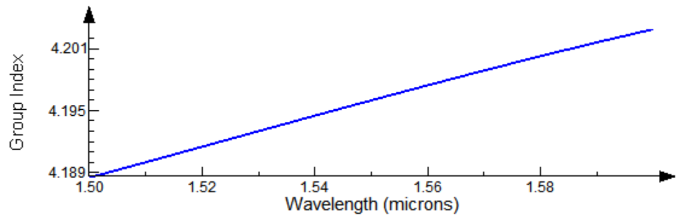


Fig. 8. Group Index vs. Wavelength (TE)

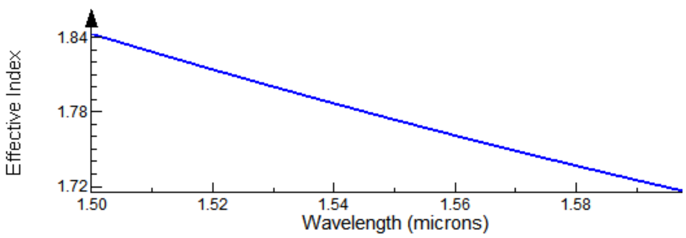


Fig. 9. Effective Index vs. Wavelength (TM)

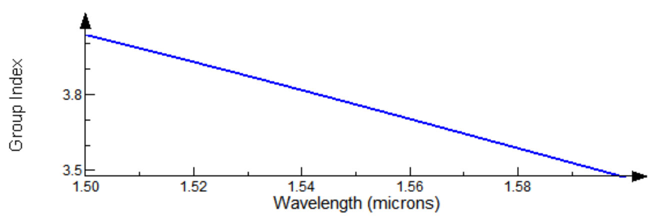


Fig. 10. Groupe Index vs. Wavelength (TM)

The compact model of the waveguide at 1.55 µ is given as per the following Taylor expression:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_o) + n_3(\lambda - \lambda_o)^2 \qquad (1)$$

Here, n1, n2, and n3 are the three fit parameters of the polynomial expression. λ_o is the free-space wavelength and is selected to be 1.55µ. λ can be any operating wavelength between 1.5µ and 1.6µ. The typical values of n1, n2, and n3 for TE polarization at λ_o =1.55µ are given as 2.44747, -1.12805, and -0.046585 respectively. The typical values of n1, n2, and n3 for TM polarization at $\lambda_o = 1.55$µ are given as 1.77436, -1.28227, and 1.80806 respectively.

1) Simulation 1: Using E-Beam Components

An unbalanced MZI is built in Interconnect using e-beam components [2]. It is built using two 50:50 broadband directional couplers and ΔL=1369.15µ. Polarization is set to TE.

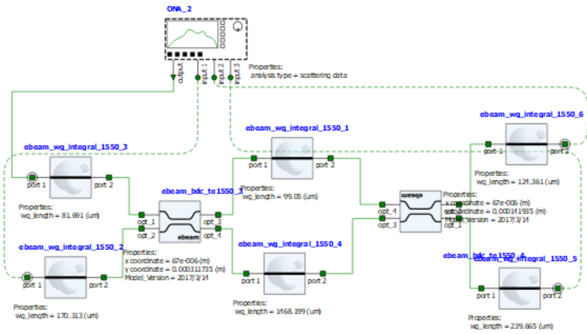


Fig. 11. Circuit Diagram

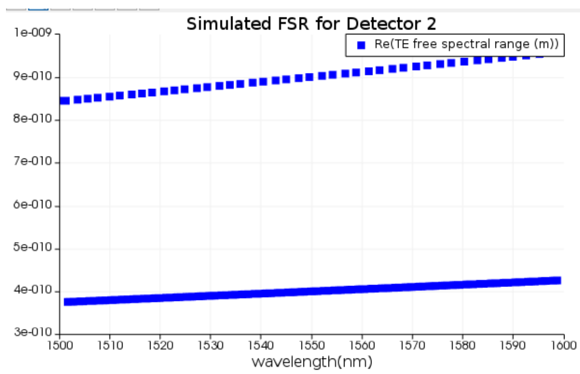


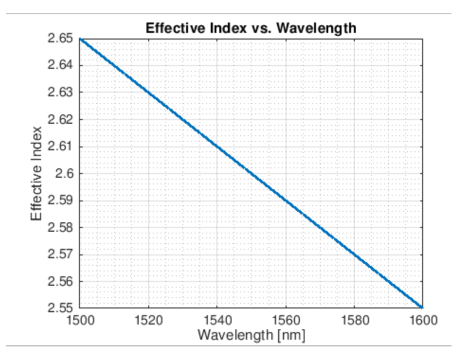
Fig. 12. Expected FSR

From Fig. 12, at λ=1.55µ, FSR obtained is 0.4 nm.

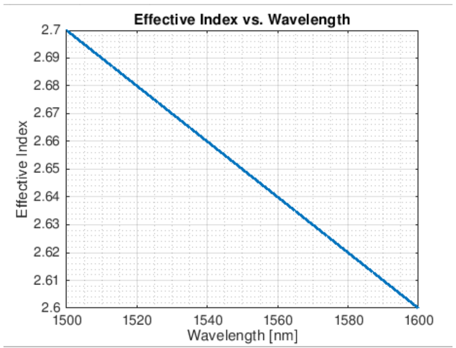
$n_g = \lambda^2 / (\Delta L\ FSR)$

$n_g = (1.55^2 * 1000)/(1369.15 * 0.4)$

$n_g = 4.39$



(a) (n1, n2 n3) = (2.6, -1, 0)



(b) (n1, n2, n3) = (2.65, -1, 0)

Fig. 13. Plot of Effective Index vs. Wavelength for two MZI Arms using Different Fit Parameters

The transfer function for MZI is as follows:

$$T = \frac{I_{out}}{I_{in}} = \frac{1}{4}\left|e^{i\beta_1 L_1}e^{-(\alpha_1 L_1)/2} + e^{i\beta_2 L_2}e^{-(\alpha_2 L_2)/2}\right|^2$$

Here, α_1 and α_2 are the loss per unit length in each MZI arm. L_1 and L_2 are the lengths of the two arms of MZI. β_1 and β_2 are related to wavelength of light and effective index of the respective arm. $\beta_1 = \frac{2\pi}{\lambda}n_1$ and $\beta_2 = \frac{2\pi}{\lambda}n_2$, where n_1 and n_2 are effective indices of the two arms.

Using the MZI Transfer Function, following transmission spectrum is obtained. Keeping loss at -3dB/cm or -0.691 cm^{-1} and using first set of fit parameters (2.6, -1, 0), following is the simulated transmission spectrum obtained for detector 2 using MATLAB.

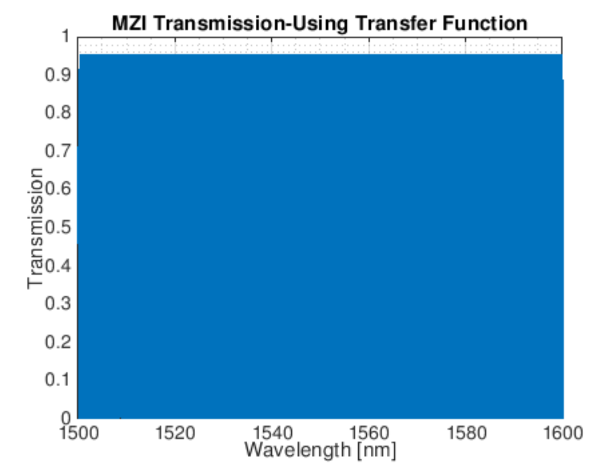


Fig. 14. Simulated Transmission Spectrum Using MZI Transfer Function for Detector 2

2) *Simulation 2: Using RIT PDK Components*

An unbalanced MZI is constructed using RIT PDK components. The MZI has ΔL=145.19µ and is designed using two 50:50 broadband directional couplers. Polarization used is TM.

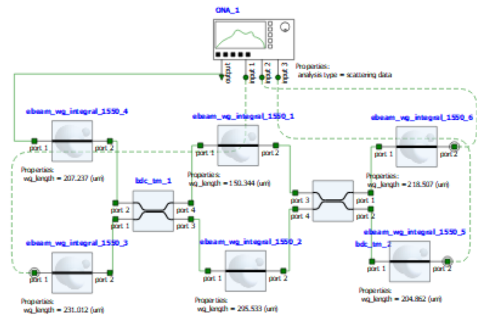


Fig. 15. Circuit Diagram ΔL=145.19µ

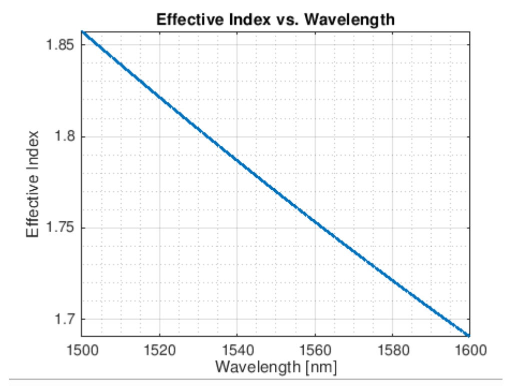


Fig. 16. Effective Index vs. Wavelength (TM Polarization)

To obtain the transmission spectrum as shown in Fig. 17, the following fit parameters have been considered:
(n1, n2, n3) = (1.77, -1.6689, 1.6)
Loss of -3dB/cm has been taken into account.

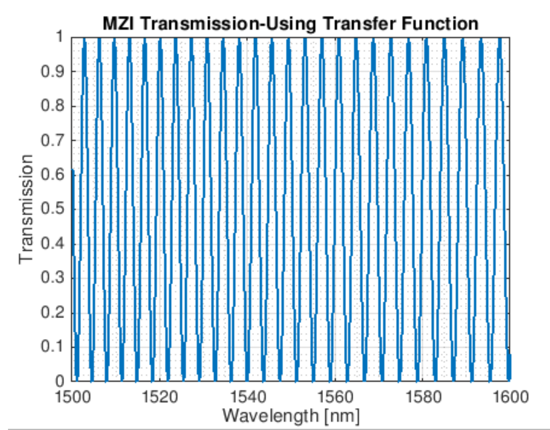


Fig. 17. MZI Transmission Spectrum ΔL=145.19µ

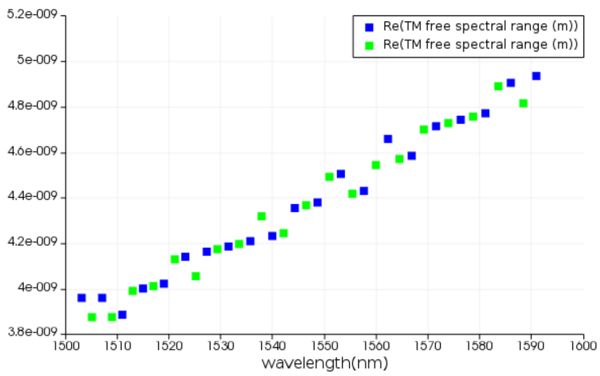


Fig. 18. FSR Obtained from Simulation Result

From simulation result, FSR is 4.5 nm at 1.55μ wavelength.

$n_g = \lambda^2 / (\Delta L\ FSR)$

$n_g = (1.55^2 * 1000)/(145.19 * 4.5)$

$n_g = 3.68$

Another similar MZI with TM polarization having ΔL=444.326μ produces the following transmission spectrum.

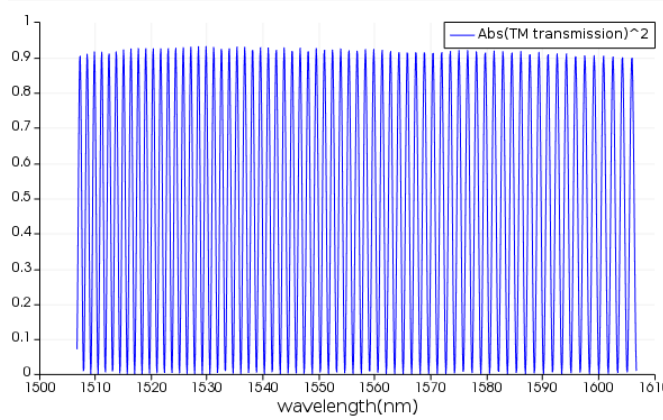


Fig. 19. MZI Transmission Spectrum ΔL=444.326μ

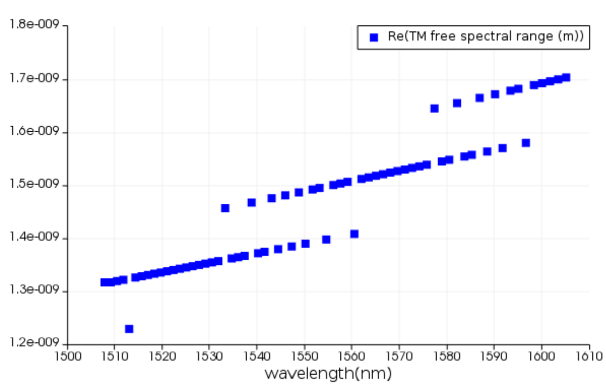


Fig. 20. FSR Obtained from Simulated Data

The simulated results provide FSR of 1.39 nm at wavelength of 1.55μ.

$n_g = \lambda^2 / (\Delta L\ FSR)$

$n_g = (1.55^2 * 1000)/(444.326 * 1.39)$

$n_g = 3.89$

3) Simulation 3: Unbalanced MZI with four rings (using e-beam components)

An unbalanced MZI is constructed using components from SiEPIC library having length difference ΔL= 288.603 µ. Since rings in SiEPIC are not functional, ring is constructed using an ideal optical waveguide coupler and a waveguide. Since the required ring radius is 10 µ, the length of the waveguide is set to 2πR=2 π*10=62.4 µ. The coupling coefficient kappa is equal to 0.04096. This coupling coefficient is extracted for 200 nm gap. Polarization is set to TE. All the rings are similar.

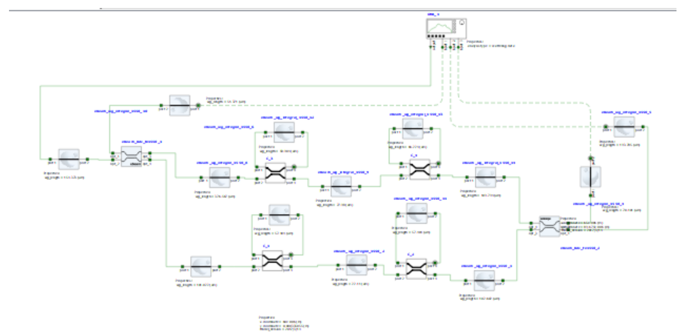


Fig. 21. Circuit Diagram for Unbalanced MZI with Four Rings

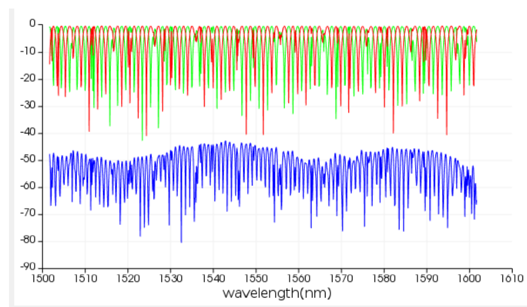


Fig. 22. Transmission Spectrum (Grating Couplers Removed)

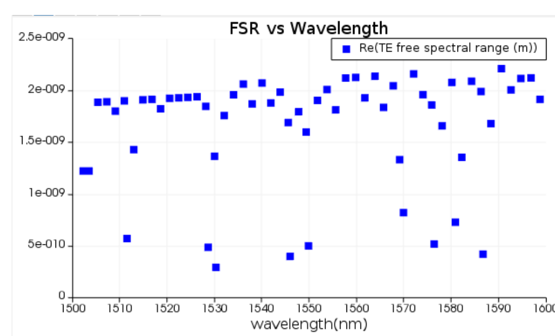


Fig. 23. FSR obtained from Simulated Results

From simulation result:

FSR at 1.55 µm wavelength ≈ 2 nm

$n_g = (1.55^2 * 1000)/(288.603 * 2)$

$n_g = 4.16$

4) Simulation 4: Unbalanced MZIs using Y-Splitters (using e-beam components)

Three unbalanced MZIs built using y-splitters are placed in series to study the overall response of the device. The length difference for the three segments of the MZI from bottom to top are 22.5µ, 39.024µ, and 20.54µ respectively. Coupling coefficient is set to 0.04096 for 200 nm gap rings. Both the rings are identical and have 10µ ring radius. The components used belong to SiEPIC library other than the directional coupler, which is an ideal component and is used to construct the ring. Polarization is set to TE.

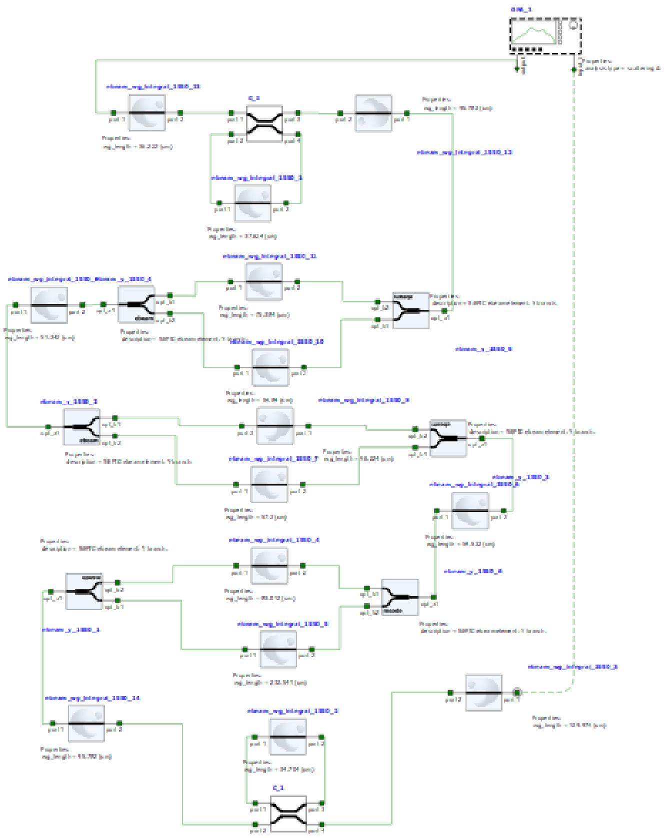


Fig. 24. Circuit Diagram

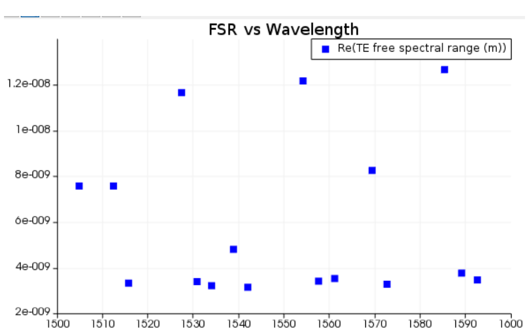


Fig. 25. FSR Obtained from Simulated Results

Due to small length difference between MZIs, not much interference is happening. Hence, FSR data is scattered and not compact. Fig. 26 shows the transmission spectrum for this device (without grating couplers).

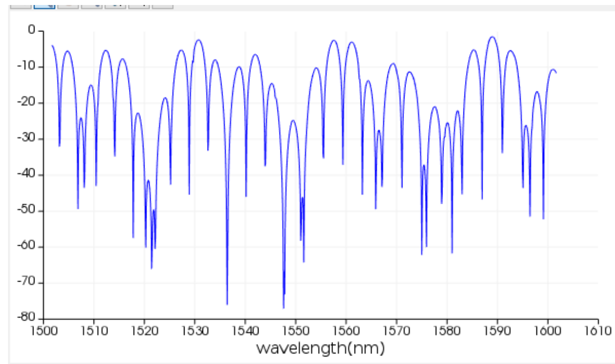


Fig. 26. Transmission Spectrum

IV. CHIP FABRICATION

A. Chip Fabricated Using E-Beam Lithography (EBL)

The first chip was fabricated at University of Washington using the NanoSOI fabrication process. The diameter of silicon wafer is 20 cm. The silicon thickness is 220 nm. The thickness of buried oxide is 2µ. 100keV electron beam is used to write the pattern on the wafer coated with resist; also known as direct-write 100 keV electron beam lithography technology. An advantage of using e-beam lithography is that very high resolution can be attained and hence it is possible to create very small feature size (≈60 nm). The wafer is diced beforehand into substrates where the dimension of each die is 25 mm x 25 mm. The wafer is first cleaned using the piranha solution (3:1 H_2SO_4:H_2O_2) for 15 minutes and rinsed with water/IPA (Isopropyl alcohol). The substrate is then spin-coated with hydrogen silsesquioxane (HSQ) and heated until the solvent evaporates. Raith EBPG 5000+ electron beam instrument, having raster step size of 5 nm, is used to generate the electron beam for pattern transferring. Post electron beam exposure, the substrates are treated with tetramethylammonium sulfate (TMAH) solution. After inspection of devices, etching process is conducted using chlorine and the resist is removed from the surface. Finally, the devices are verified under scanning electron microscope (SEM) to verify the fidelity of pattern and etch quality.

B. Chip fabricated at RIT using Photolithography

The following three stages were involved in the chip-making process at RIT:

Stage 1) Mask Making and Wafer Preparation

The wafer is RCA cleaned to remove any contaminants. The DWL 66+ laser lithography system is a pattern generator used for mask making and direct writing. The exposure wavelength is 405 nm.

Stage II) Photolithography

The main steps involved in Photolithography cycle are:

a. *Carbon, BARC and photoresist coating:* BARC (Icon 7a) is coated on the wafer and subjected to 2000 rpm for 30s. Bake is performed at 170° C for 60s. The photoresist layer OiR 620 is coated on the wafer and again subjected to 2000 rpm for 30s. Bake is performed at 100° C for 60s.

b. *I-line Exposure of mask onto wafer:* The wafer is 150 mm thick. Exposure is done in the ASML stepper which is a 5x stepper. The features on photomask are reduced 5x down. Exposure is done using mercury i-line at 365 nm.

c. *Development of the resist image:* The wafer is developed for 90 s using CD-26. This dissolves the exposed OiR 620. Bake is performed at 145º C for 45s. Finally, inspection is done under the microscope to verify the waveguide thickness and gaps. The BARC is removed to get a clean silicon surface for etching.

Stage III) Etching

BARC etching is done using oxygen plasma at power of 30W for 30s at 5 s.c.c.m. Post that, shallow etching is performed in the STS etcher using plasma made from C_4F_8, SF_6, O_2 and Ar. The O_2 plasma is used to strip the photoresist.

V. E-BEAM EXPERIMENTAL DATA

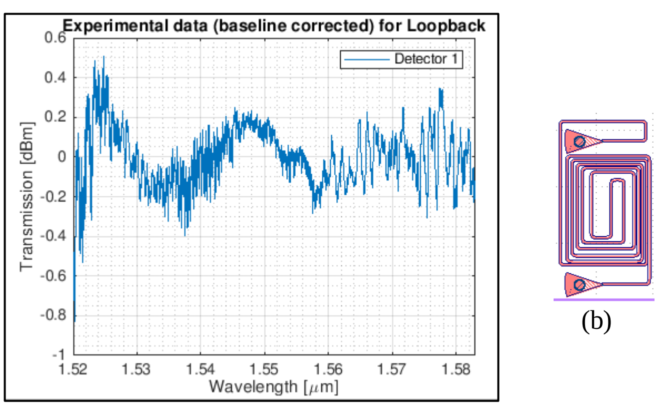
The E-Beam design was laid in KLayout on a grid having dimension of 605 µm (width) X 410 µm (height). The grating couplers for TE polarization were placed facing right. The grating couplers for TM polarization were placed facing left. The spacing between each grating coupler is 127µ. Counting from top to bottom, the second fiber in the fiber array serves as the input. The first, third, and fourth fiber in the array are be used for output.

A. E-beam Devices using TE polarization

Baseline fit is done using polynomial order 4 or 5 for all the devices. Baseline fit is done essentially to remove the grating coupler response and flatten the response of the devices.

1) Device 1: Loopback

The length of loopback is 1784.73 µm and has a bend radius of 5 µ.

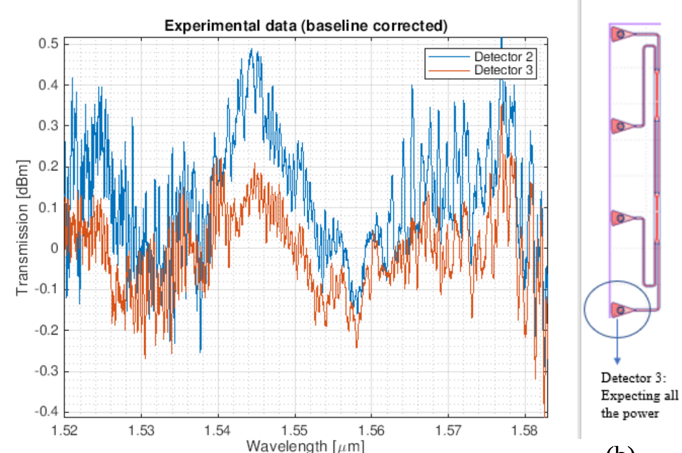


(a) (b)

Fig. 27. (a) Baseline Corrected Experimental Data (b) Loopback Structure

2) Device 2: Balanced MZI Using Two 50:50 Broadband Directional Couplers

It can be observed from Fig. 28(a) that although the MZI is balanced, it is slightly unbalanced. Ideally, no power is expected in detector 2 because of destructive interference.

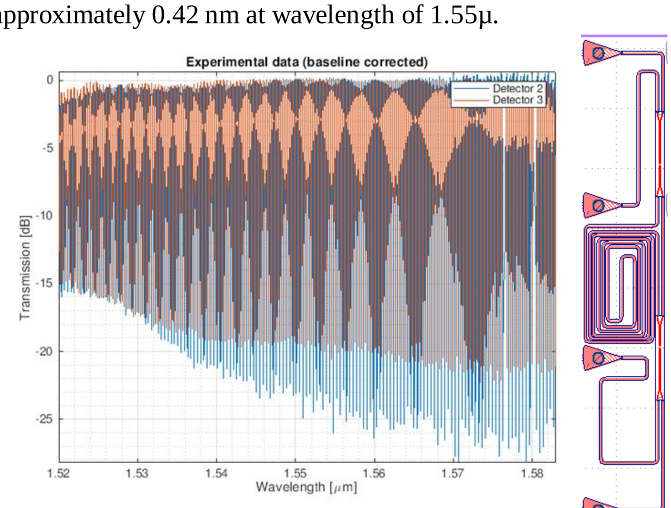


(a) (b)

Fig. 28. (a) Baseline Corrected Experimental Data (b) Balanced MZI Structure

3) Device 3: Unbalanced MZI Using Two 50:50 Broadband Directional Couplers

The length difference ΔL between the two arms of MZI is 1369.15µ. Minimum allowable FSR limit is 0.1 nm, which is limited by the laser's resolution. The reason for selecting such big length difference is to attempt to make the FSR as narrow as possible. Here as observed from Fig. 30(a), FSR is approximately 0.42 nm at wavelength of 1.55µ.

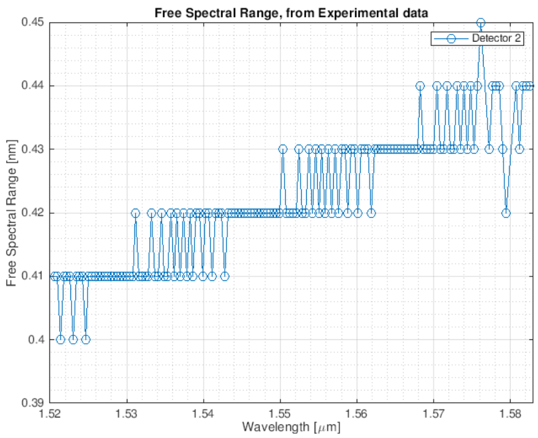


(a) (b)

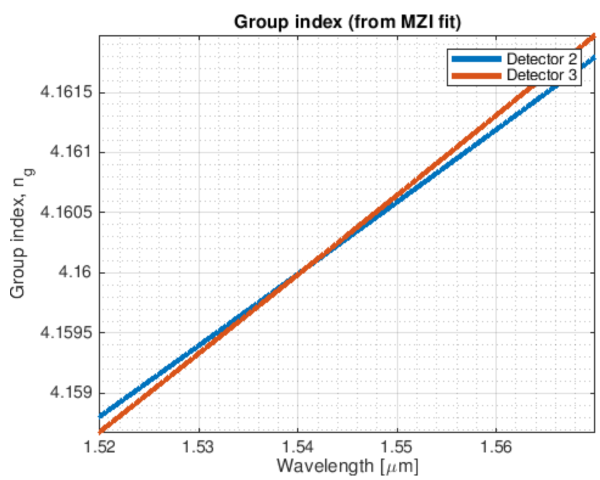
Fig. 29. (a) Baseline Corrected Experimental Data (b) Unbalanced MZI Design

TABLE III
FIT PARAMETERS AND GROUP INDEX VALUES

DETECTOR 2	DETECTOR 3
Fit parameters (n1, n2, n3): (2.6, -1, 0)	Fit parameters (n1, n2, n3): (2.65, -1, 0)
n_g=4.1603 @ λ=1.545 µm	n_g=4.1603 @ λ=1.545 µm



(a)

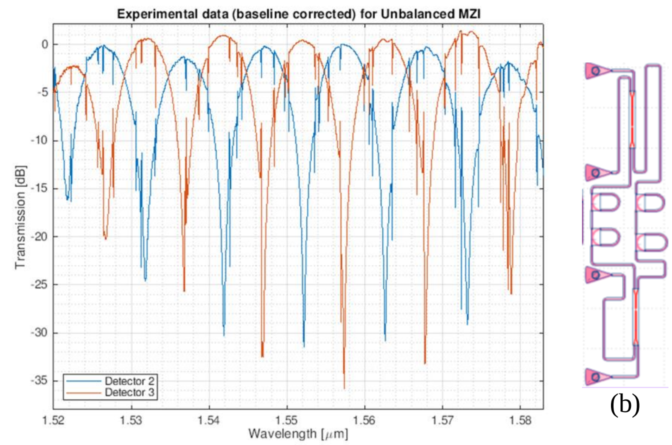


(b)

Fig. 30. (a) FSR Obtained from Experimental Data (b) Group Index

4) Device 4: Unbalanced MZI With Four Rings

Following parameters have been considered in this design:
MZI Length Difference ΔL= 288.603 µ
Rings Gap: 200 nm
Rings Radius: 10 µ
Ring Circumference=2πR=2 π*10=62.4 µ



(a) (b)

Fig. 31. (a) Baseline Corrected Experimental Data (b) Unbalanced MZI Design

Group Indices for this design could not be found using the MATLAB script.

5) Device 5: Unbalanced MZIs Using Y-Splitters and Rings

Following design parameters have been for this device:
MZI1 Length Difference ΔL_1= 22.5 µ
MZI2 Length Difference ΔL_2= 39.024 µ
MZI3 Length Difference ΔL_3= 20.54 µ
Rings Gap: 200 nm
Rings Radius: 10 µ

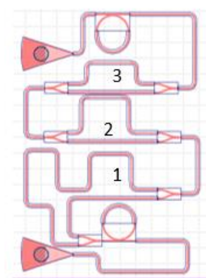


Fig. 32. Device Design

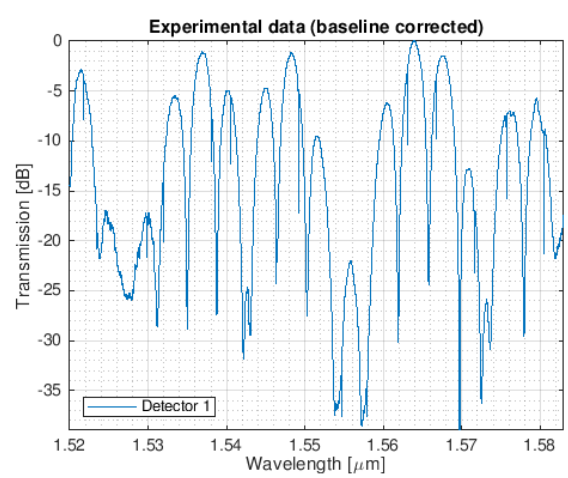
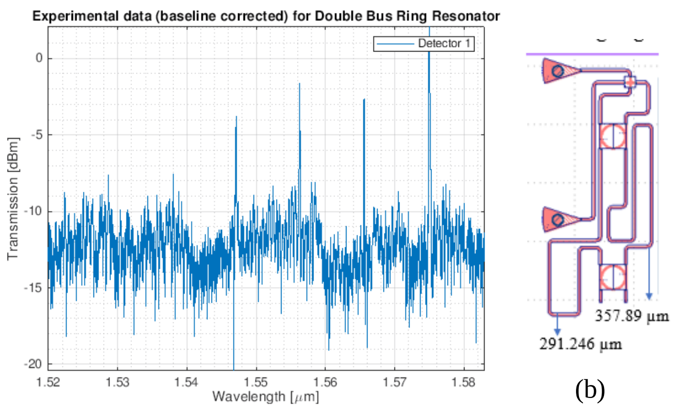


Fig. 33. Baseline Corrected Data

MZIs length difference is small, hence did not get enough oscillations. Therefore, FSR and group indices could not be calculated using MATLAB scripts. However, simulation results are shown in the simulation section.

6) Device 6: Double-Bus Ring Resonators

Transmission in Detector 1
(with grating couplers) < -60 dBm
Length of arm1= 291.246µ
Length of arm2=357.89µ
Radius of both the rings=10µ
Gap for both rings=0.2µ

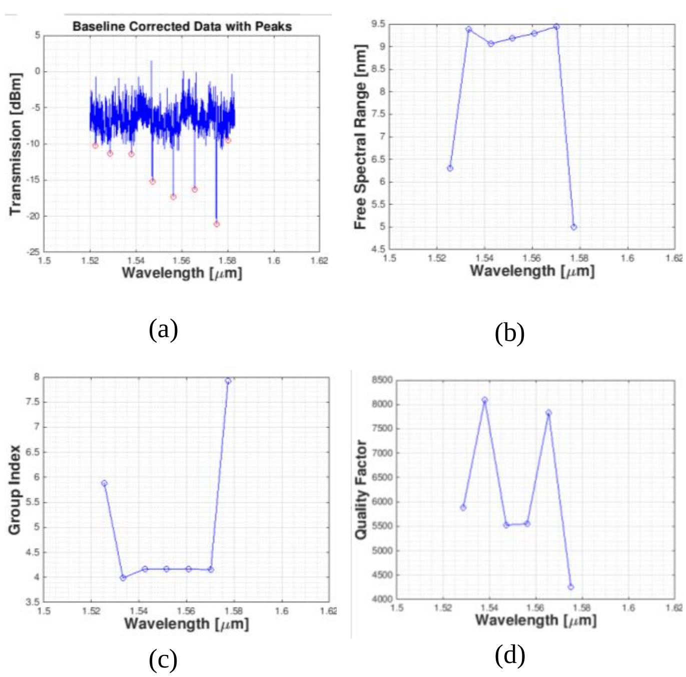


(a) (b)

Fig. 34. (a) Baseline Corrected Data (b) Device Structure

The minima peaks, FSR, Quality Factor and Group Index values are derived using negative thru. Following parameters are considered for finding minima peaks:

Minimum resonance separation: 9 nm
Minimum resonance depth: 4 dBm



(a) (b) (c) (d)

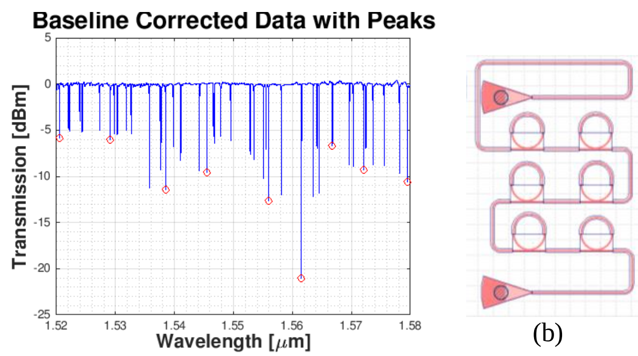
Fig. 35. (a) Baseline Corrected Data with Thru Reversed (b) FSR Obtained from Experimental Data (c) Group Index (d) Quality Factor

7) Device 7: Loopback Using Six Single-Bus Ring Resonators

Following are the design parameters used for all the six rings:
Ring Gap: 0.2 µm
Ring Radius: 10 µm
Ring Circumference: 62.8 µm

It must be noted that even though the rings may visually appear to be of different radii, they are all the same radii and have the same gap size.

Loopback length excluding rings: 531.68 µm

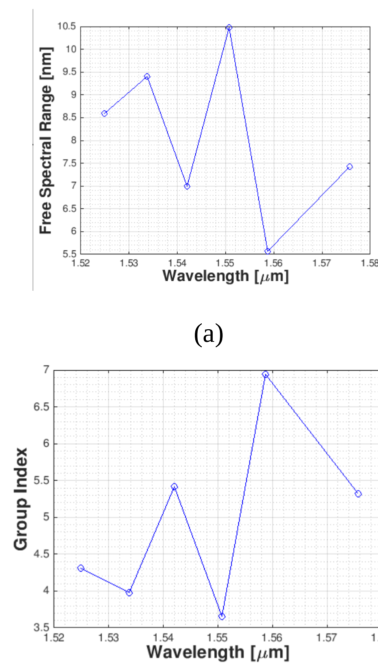


(a) (b)

Fig. 36. (a) Baseline Corrected Data (b) Device Design

The following parameters are considered to find minima peaks:

Minimum resonance separation: 5 nm
Minimum resonance depth: 4 dBm



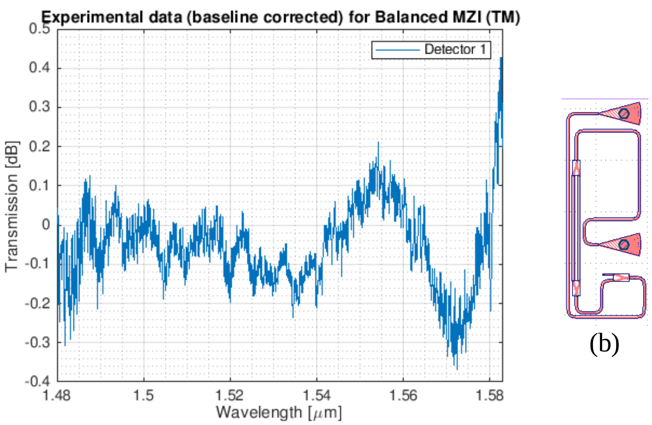
(a) (b)

Fig. 37. (a) FSR Obtained from Experimental Data (b) Group Index

B. E-beam devices using TM polarization

8) Device 8: Balanced MZI Using Y-Splitters

Balance MZI is constructed using two y-splitters and ΔL=0µ. It can be observed that MZI is fairly balanced; however, at bigger wavelengths after 1.55 µ data tends to become bad. The bend radius is 10µ.



(a) (b)

Fig. 38. (a) Baseline Corrected Data (b) Balanced MZI Design

VI. RIT EXPERIMENTAL DATA

A. RIT Test Procedure

The layout for RIT chip is designed in KLayout on a 2050µx2050µ grid. The photonic chip fabricated at RIT using photolithography is tested using the automated probe station at Center of Photonic Communications. A tunable laser (ECL) provided by Keysight 8164B is used as laser source. Power set in the laser is 6 dBm. The resolution size is set to 1 ps. The laser sweeps from 1.5µ to 1.6µ. A single-mode polarization maintaining (PM) fiber is used to shoot light from laser into the fiber array, which is oriented close enough to the wafer. The devices are then scanned one by one [3]. To keep the design compatible with the E-beam design, laser input is provided to fiber 2 in the array and the remaining three fibers in the array are used for detecting output power. The distance between each grating coupler (within a cell) in the design is 127µ. The pitch of grating couplers is 1.25µ. The CSV and .mat files received from test results are finally processed in MATLAB for further study.

B. Devices on RIT chip

All devices on the RIT chip use only TM polarization. An advantage of using TM polarization is that its mode effective index is lower; hence coupling of light happens easily even over bigger gaps size. The bend radius for all the devices on RIT chip is 15µ.

1) Device 1: Loopback Using Single-Bus Ring Resonator

The following parameters are considered while designing the device:

Ring Radius R= 15 µm

Ring Circumference=2πR=94.2 µm

Waveguide Length (excluding ring circumference) =258.034µm

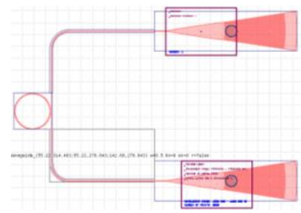
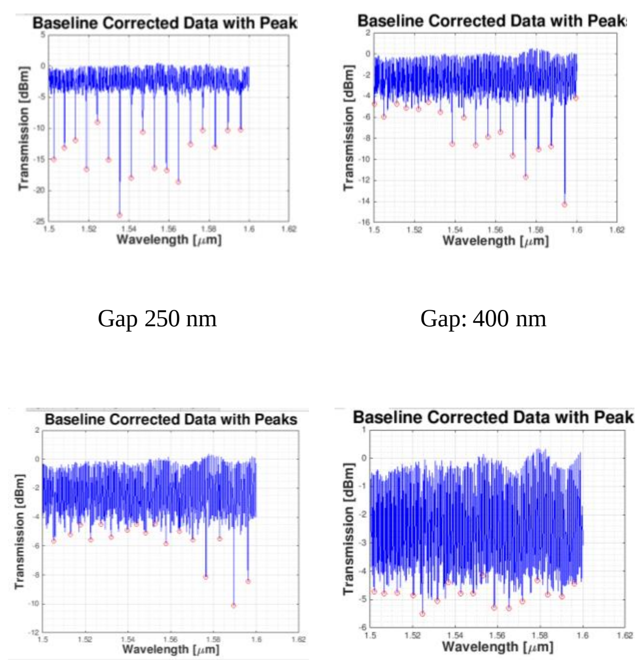


Fig. 39. Single-bus Ring Resonator

Usually for TM devices, 700 nm gap for rings is considered ideal; however, the amount of loss experienced in the automated test system is quite high. It is observed from the following data that 250 nm gap can be considered appropriate since pretty good resonances are observed at this gap. However, as the gap is increased, data begins to mess up. To find minima peaks, following parameters are considered:

Minimum resonance separation: 5 nm

Minimum resonance depth: 4 dBm



Gap 250 nm — Gap: 400 nm

Gap: 500 nm — Gap: 700 nm

Fig. 40. Baseline Corrected Data for Different Gaps Size

It can be observed that due to high amount of reflections light is travelling back and forth which is causing interference to happen. This is causing FSR to behave unexpectedly, particularly for bigger gaps size.

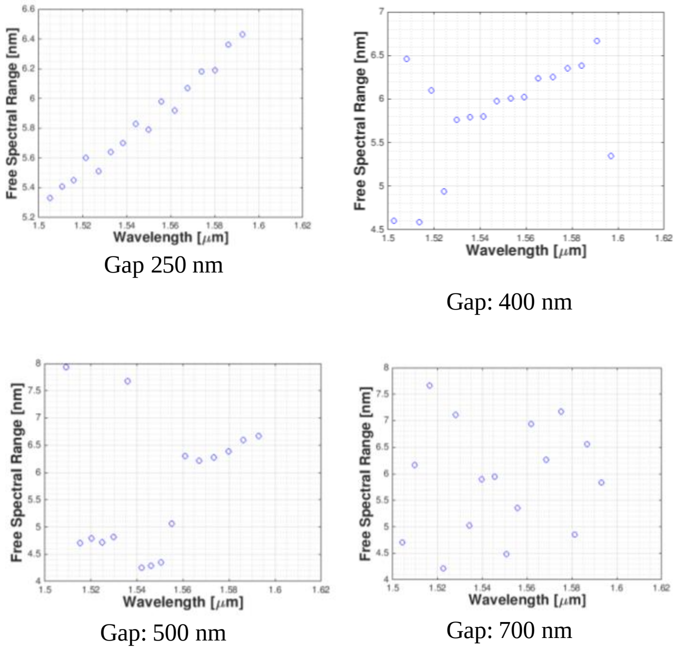


Fig. 41. FSR for Rings with Different Gaps Size

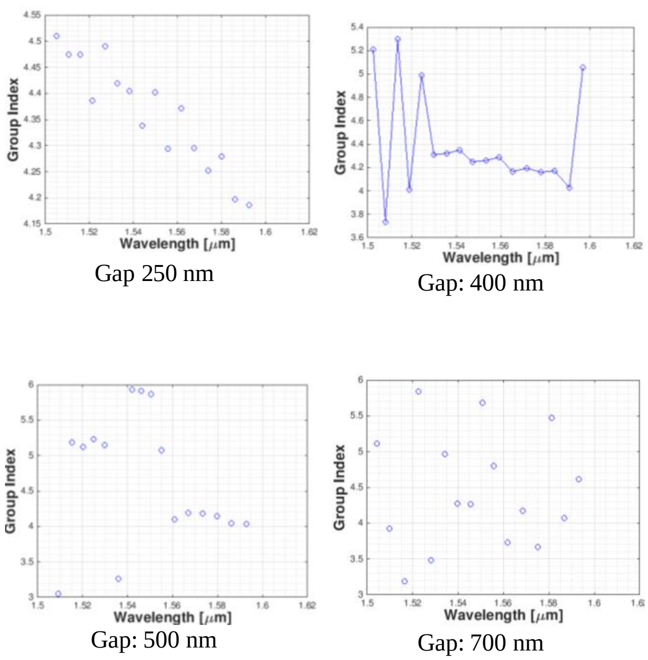


Fig. 42. Group Indices Obtained for Different Gaps Size

For bigger gaps size particularly for 700 nm gap, system acts extremely noisy. The amount of reflections increase considerably, and the quality of resonances become very poor.

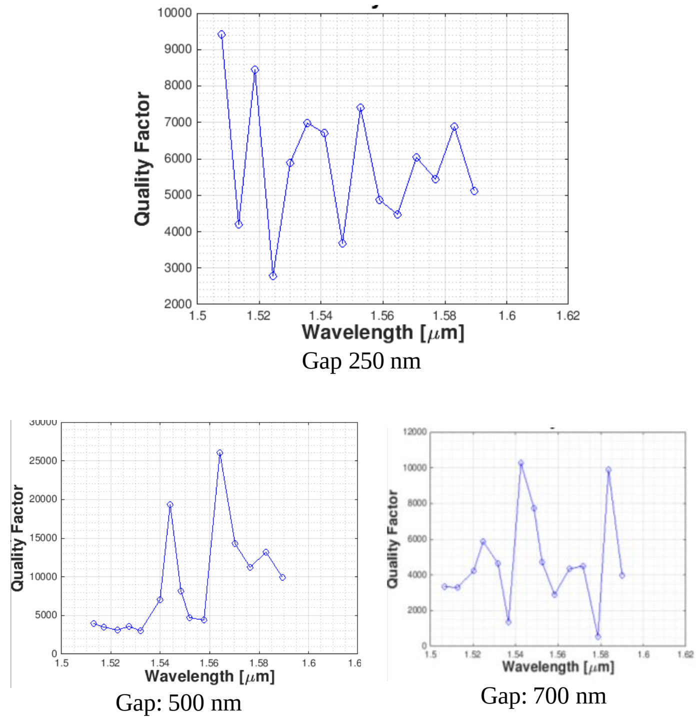


Fig. 43. Quality Factor for Different Gaps Size

2) *Device 2: Balanced MZI Using Two 50:50 Broadband Directional Couplers*

A balanced MZI having length difference ΔL=0μ is created using two 50:50 broadband directional couplers. However, it can be observed good amount of transmission is observed in both the arms. The MZI is not perfectly balanced and a possible reason for this could be fabrication defect.

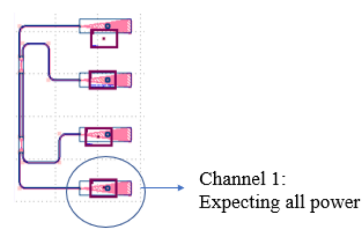


Fig. 44. Balanced MZI

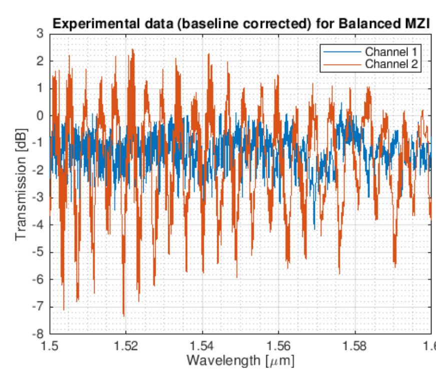


Fig. 45. Baseline Corrected Data for Balanced MZI

3) Device 3: Unbalanced MZI Using Two 50:50 Broadband Directional Couplers

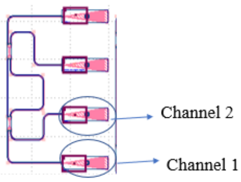


Fig. 46. Unbalanced MZI

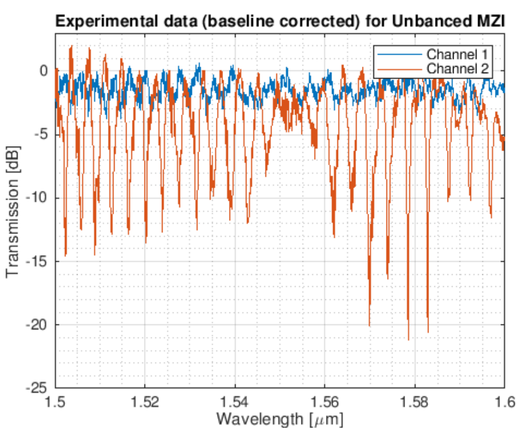
Data for two MZIs has been provided here:

1. ΔL= 145.19 µm
2. ΔL= 444.326 µm

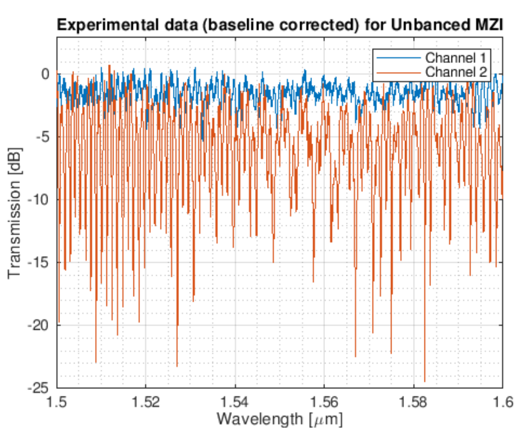
As ΔL is increased, FSR is expected to become smaller. As observed from Fig. 48, when ΔL= is increased to 444.326 µm, FSR is reduced by almost 3 nm. Mathematically, FSR is given as:

$$FSR = \lambda^2/(\Delta L n_g)$$

Here, λ is the operating wavelength and n_g is the group index.

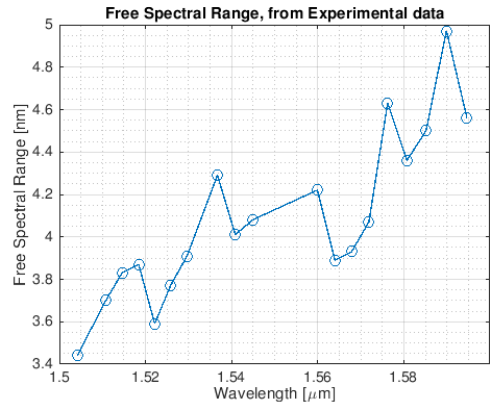


ΔL= 145.19 µm

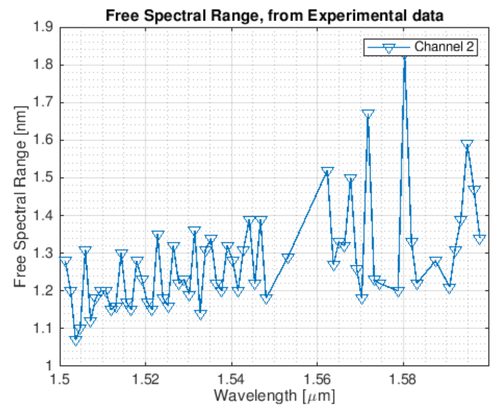


ΔL= 444.326 µm

Fig. 47. Baseline Corrected Data for MZIs with Different ΔL



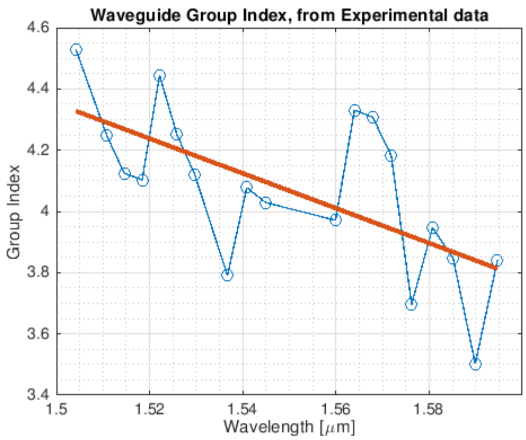
ΔL= 145.19 µm



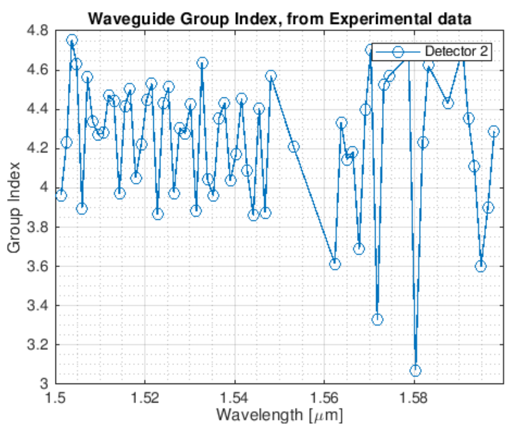
ΔL= 444.326 µm

Fig. 48. FSR for MZIs with Different ΔL

The fit parameters obtained here are (n1, n2, n3) = (1.77, -1.6689, 1.6). Goodness of fit is 61.48%. Channel 1 is fairly balanced (no oscillations); hence FSR or group index cannot be derived for it. The FSR data and Group Index is obtained for channel 2.



ΔL= 145.19 µm



ΔL= 444.326 µm

Fig. 49. Group Indices for Unbalanced MZIs with Different ΔL

4) *Device 4: Unbalanced MZI Using Two Y-Splitters and a Ring*

Two MZIs are constructed using Y-splitters having similar length difference but different ring gaps. First MZI has a ring gap of 400nm and second MZI has a ring gap of 250 nm.

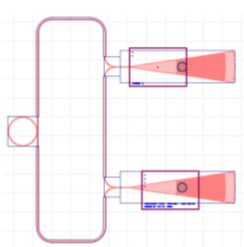
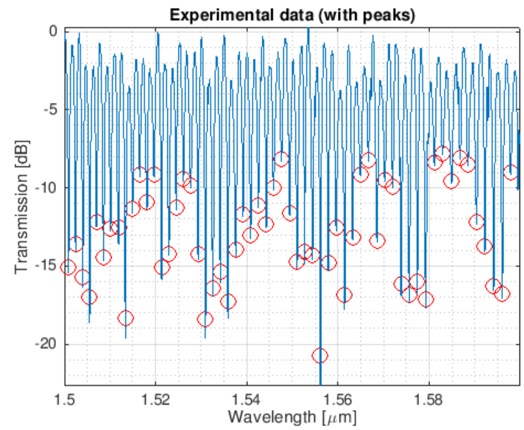
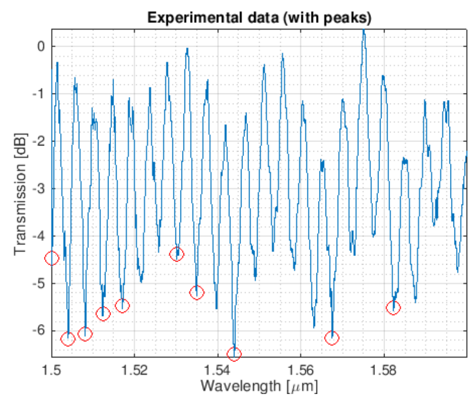


Fig. 50. Unbalanced MZI using Y-Splitters and a Ring

Fig. 51. shows the transmission spectra for both the MZIs.

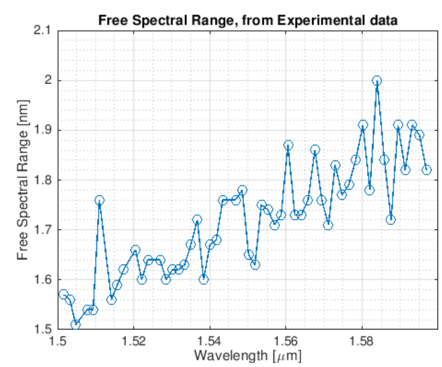


ΔL= 300.168 µm; Ring Gap=400 nm



ΔL= 300.168 µm; Ring Gap=250 nm

Fig. 51. Experimental Data for Unbalanced MZIs with Rings of Different Gaps Size



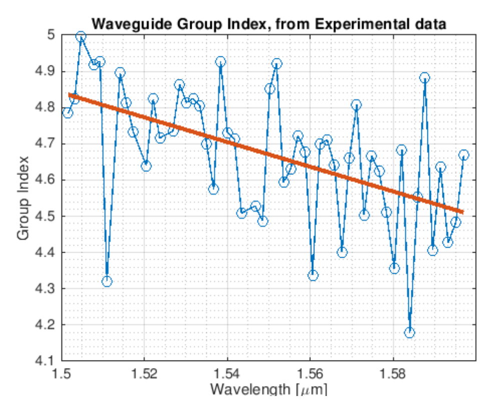


Fig. 52. FSR and Group Index for ΔL= 300.168 µm; Ring Gap=400 nm

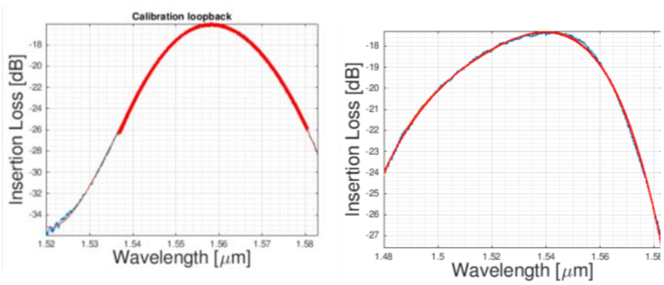
For MZI having ring radius of 0.4µ, the fit parameters obtained are (n1, n2, n3): (1.769, -1.8767, 1.1035). Goodness of fit obtained is 0.81437.

For MZI having ring radius of 0.25µ, the group indices could not be found using MATLAB scripts possibly due to a smaller number of minima peaks.

VII. ANALYSIS

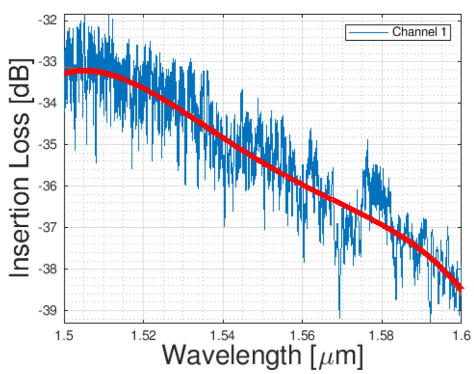
The baseline fit for all the devices is done using polynomial order 4 or 5.

A. Insertion Loss Analysis



a) TE Polarization *b) TM Polarization*

Fig. 53. Insertion Loss Measured in E-Beam Devices



TM Polarization

Fig. 54. Insertion Loss Measured in RIT Devices

As observed from Fig. 54(a), minimum loss for E-Beam devices is observed at 1.56µ for TE polarization, which is close to 17dB. Loss increases at very low or at very high wavelengths.

TABLE IV
INSERTION LOSS COMPARISON OF E-BEAM AND RIT CHIP DEVICES

	E-beam devices		RIT Devices
Wavelength [µ]	TE Loss [dB]	TM Loss [dB]	TM Loss [dB]
1.52	35	18.2	33.5
1.55	17.9	17.9	35
1.58	26	27.5	37

B. Analysis of MZI Data

As observed from Table V, FSR decreases as ΔL increases. Group index for MZI4 having TE polarization is as expected. However, for MZIs having TM polarization, group index obtained is bit higher. The expected group index for devices having TM polarization is between 3 and 4.

C. Analysis of Single-Bus Ring Resonators

TABLE V
COMPARISON OF MZI PERFORMANCE DATA

	RIT Chip			E-Beam Chip
Parameters	MZI1	MZI2	MZI3	MZI4
Polarization ΔL [µ]	TM 145.19	TM 300.168	TM 444.326	TE 1369.15
FSR [nm] @ 1.55 µ	4.1	1.62	1.18	0.42
n_g @ 1.55 µ	4	4.7077	4.2	4.1603
Fit Parameters (n1, n2, n3)	(1.77, -1.6689, 1.6)	1.769, -1.8767, 1.1035	(1.77, -1.6689, 1.6)	(2.65, -1, 0) (2.6, -1, 0)
Fit Goodness	0.6148	0.81437	0.6148	0.69041
Dispersion [ps/nm-km]	-18973.3	-15476.62	-6344.99	673.8243

TABLE V
COMPARISON OF RINGS PERFORMANCE DATA

	RIT Chip [TM Polarization]			
Parameters	Ring1	Ring2	Ring3	Ring4
Gap [nm]	250	400	500	700
Ring Radius [µ]	15	15	15	15
FSR [nm] @ 1.55 µ	5.8	6	4.48	4.5
n_g @ 1.55 µ	4.4	4.22	6	5.55
Quality Factor	5000	--	5000	5000
Quality of Resonances	Good	Average	Bad/ Noisy	Very bad/ Extremely Noisy

Highest quality factor (>25000) is obtained at λ≈1.57 µm for ring having gap of 500 nm. Group indices obtained are unexpectedly high.

D. *Simulation Results vs. Experimental Data*

1)

TABLE VI
COMPARISON OF UNBALANCED MZI ΔL=1369.15μ (TE)

	Simulation Results	E-Beam Experimental Data
FSR @ 1.55μ [nm]	0.4	0.42
n_g @ 1.55μ	4.39	4.1605

Simulated results are in close proximity with the experimental data. Similar MZI having ΔL=1130.94μ on RIT chip did not work.

2)

TABLE VII
COMPARISON OF UNBALANCED MZI (TM)

		Simulation Results	RIT Experimental Data
ΔL=145.19μ	FSR @ 1.55μ [nm]	4.5	4.1
	n_g @ 1.55μ	3.68	4
ΔL=444.326μ	FSR @ 1.55μ [nm]	1.39	1.2
	n_g @ 1.55μ	3.89	4.4

Table VII shows that group index obtained via experimental data is higher.

3)

TABLE VIII
COMPARISON OF UNBALANCED MZI WITH FOUR RINGS
ΔL=288.603μ (TE)

	Simulation Results	E-Beam Experimental Data
FSR @ 1.55μ [nm]	2	FSR and Group Indices not found.
n_g @ 1.55μ	4.16	

As stated in Table VIII, FSR and group index values cannot be obtained from e-beam experimental data using MATLAB script possibly due to smaller number of minima peaks.

E. *Corner Analysis*

Corner analysis is performed considering only the width and height variations in the waveguide to account for manufacturing variability. Although the waveguide is expected to be perfectly 500nmx220nm in dimension, it is likely that during fabrication waveguide dimensions may slightly alter. This can result in overall change in effective index of the waveguide, which can lead to interference/phase change. As a result of this, transmission will be impacted. There are two possible ways to study/analyze the impact of manufacturing variability on transmission data:

1. Monte-Carlo Analysis
2. Corner Analysis

Here, Corner analysis is performed for MZI having ΔL=145.19μ with TM polarization. Following four corners are considered along with the ideal waveguide dimension:

Ideal- Width: 500nm, Height 220nm
Corner 1- Width: 510nm, Height 223.1nm
Corner 2- Width: 510nm, Height 215.3nm
Corner 3- Width: 470nm, Height: 215.3nm
Corner 4- Width: 470nm, Height: 223.1nm

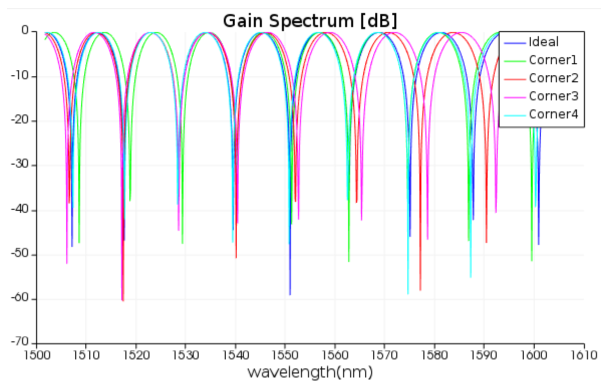


Fig. 55. Gain Spectrum for Different Waveguide Dimensions

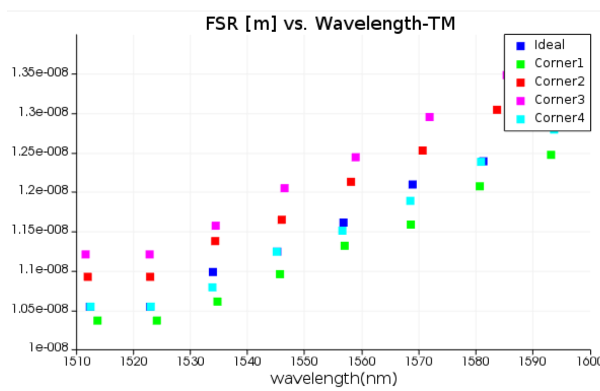


Fig. 56. FSR for Waveguides with Different Dimensions as Represented by Different Corners

It is observed that the data for ideal waveguide (500nmx220nm) falls between the data obtained for the four corners. Hence. corner analysis test can be considered pass.

TABLE IX
EFFECTIVE AND GROUP INDICES VARIATIONS FOR DIFFERENT WAVEGUIDE DIMENSIONS FOR TM POLARIZATION AT 1.55µ

	Dimension	Effective Index	Group Index
Ideal	500nmx220nm	1.77	3.76
Corner 1	510nmx223.1nm	1.8	3.88
Corner 2	510nmx215.3nm	1.75	3.62
Corner 3	470nmx215.3nm	1.72	3.52
Corner 4	470nmx223.1nm	1.75	3.78

TABLE X
EFFECTIVE AND GROUP INDICES VARIATIONS FOR DIFFERENT WAVEGUIDE DIMENSIONS FOR TE POLARIZATION AT 1.55µ

	Dimension	Effective Index	Group Index
Ideal	500nmx220nm	2.448	4.1962
Corner 1	510nmx223.1nm	2.475	4.1818
Corner 2	510nmx215.3nm	2.455	4.1745
Corner 3	470nmx215.3nm	2.378	4.2458
Corner 4	470nmx223.1nm	2.415	4.2542

To find the effective and group indices as shown in Table IX and Table X, simulations are run in Lumerical Mode Solutions software for different waveguide width and height. The effective index and group index for each waveguide is noted at 1.55µ wavelength.

F. *Observations*

1. One peculiar point observed regarding the unbalanced MZI built using 50:50 broadband directional couplers on the RIT chip is that only one of the arms seems to be oscillating. The other arm doesn't oscillate (no interference) and appears balanced.
2. Amount of insertion loss experienced in RIT devices is much higher than the loss experienced in devices on the e-beam chip. Hence most devices having longer waveguide lengths did not work.

VIII. CONCLUSION

We analyzed performance data of e-beam and RIT chip devices, particularly for FSR and group index. MZI performance was analyzed for different values of ΔL. Relationship between FSR and ΔL was studied by analyzing and comparing the experimental and simulation results. Test results obtained for single-bus ring resonators were processed to extract their quality factor and the impact of gap size on performance of the device was examined.

Abbreviations and Acronyms

BOX: Buried Oxide
ECL: External Cavity Laser
RCA: Radio Corporation of America
SCCM: Standard Cubic centimeter per minute
SOI: Silicon-on-insulator
TE: Transverse Electric
TM: Transverse Magnetic

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